

HW4

1. if/else if checks conditions in order and runs one matching branch. A series of

separate if statements evaluate each condition and may run multiple blocks

3. A flag is a variable used to signal a condition or event. It starts with a default value and is changed when something happens, the other code checks it.

5. && returns true only if both conditions are true

It uses short-circuit evaluation: if the first condition is false, the second one is not checked.

9. relational

17. lower

30. A) t=1 B) t=0 C) t=0 D) t=1

31. if (x%2==0) {

y = x / 2
}

33. double commission

int sales

if (sales <= 10000) {

commission = 0.10;

} else if (sales <= 15000) {

commission = 0.15;

} else {

commission = 0.20;
}

43. T

60. ① the else is before else if

② 90~100 is invalid in this program

65. an extra :, which should be deleted.

Programming challenges

19.

```
programs > HW4_PC19.cpp
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      double wavelength;
6      cout << "The wavelength of the electromagnetic wave: ";
7      cin >> wavelength;
8      if (wavelength >= 1E-2)
9      {
10         cout << "Radio wave \n";
11     }
12     else if (wavelength >= 1E-3)
13     {
14         cout << "Microwave \n";
15     }
16     else if (wavelength >= 7E-7)
17     {
18         cout << "Infrared wave \n";
19     }
20     else if (wavelength >= 4E-7)
21     {
22         cout << "Visible light wave \n";
23     }
24     else if (wavelength >= 1E-8)
25     {
26         cout << "Ultraviolet wave \n";
27     }
28     else if (wavelength >= 1E-11)
29     {
30         cout << "X-ray \n";
31     }
32     else
33     {
34         cout << "Gamma ray \n";
35     }
36 }
```

```
(base) PS C:\Users\brack\Desktop\C++> & 'c:\Users\brack\Desktop\C++\Engine-In-dlvtzwnb.m5i' '--stdout=Microsoft-MIEngine-0
Users\brack\.vscode\mingw64\bin\gdb.exe' '--interpret
The wavelength of the electromagnetic wave: 1E-10
X-ray
(base) PS C:\Users\brack\Desktop\C++>
```